

Ron has a bag of five marbles: 1 red marble, 1 yellow marble, 1 green marble, 1 blue marble, and 1 white marble. Suppose Ron picks one marble out of the bag randomly, replaces it, and then picks another marble out. The outcome table represents the possible outcomes.

		Pick 2				
		Red	Yellow	Green	Blue	White
Pick 1	Red	(R,R)	(R,Y)	(R,G)	(R,B)	(R,W)
	Yellow	(Y,R)	(Y,Y)	(Y,G)	(Y,B)	(Y, W)
	Green	(G,R)	(G,Y)	(G,G)	(G,B)	(G,W)
	Blue	(B,R)	(B,Y)	(B,G)	(B,B)	(B,W)
	White	(W,R)	(W,Y)	(W,G)	(W,B)	(W,W)

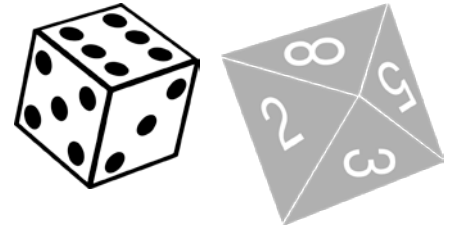
- Complete the grey row to represent the possible outcomes from choosing a second marble.
- Complete the remainder of the outcome table.
- Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Randomly choosing a marble	5	5	25

- How many outcomes are in Event A = choosing yellow both times?

$$\frac{1}{25}$$

Antonia is playing a game with two of her classmates. For each player's turn, a fair six-sided die is rolled and then a fair eight-sided die is rolled.



5. How many outcomes are in the sample space when the six-sided die is rolled?

6

6. How many outcomes are in the sample space when the eight-sided die is rolled?

8

7. How many outcomes are in the sample space when the six-sided die is rolled, then the eight-sided die is rolled?

48

8. Explain which method you would choose to determine the sample space for this random process: a list, outcome table, or tree diagram. **Answers may vary.**

Table because it organizes all the possible outcomes.

9. Use your chosen method from question 8 to determine the sample space.

		Roll 2							
Roll 1		1	2	3	4	5	6	7	8
	1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	(1,7)	(1,8)
	2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)	(2,7)	(2,8)
	3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)	(3,7)	(3,8)
	4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)	(4,7)	(4,8)
	5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)	(5,7)	(5,8)
	6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)	(6,7)	(6,8)

10. Write out the sample space for one player's turn.

Sample space: {11, 12, 13, 14, 15, 16, 17, 18, 21, 22, 23, 24, 25, 26, 27, 28, 31, 32, 33, 34, 35, 36, 37, 38, 41, 42, 43, 44, 45, 46, 47, 48, 51, 52, 53, 54, 55, 56, 57, 58, 61, 62, 63, 64, 65, 66, 67, 68}